

## OPERATIONS SUMMARY

Total Operations: 20

Part Number: 12

## Operation 1: Drilling

|                   |             |                   |                  |
|-------------------|-------------|-------------------|------------------|
| Material:         | steel       | Machine:          | Drilling machine |
| Man Hours/Unit:   | 1800        | Duty Category:    | heavy            |
| Setup Time (min): | 25          | Cycle Time (min): | 120              |
| Diameter:         | 22          | Length:           | 65               |
| Shape:            | rectangular |                   |                  |

## Operation 2: Turning

|                   |           |                   |        |
|-------------------|-----------|-------------------|--------|
| Material:         | aluminium | Machine:          | CNC    |
| Man Hours/Unit:   | 2400      | Duty Category:    | medium |
| Setup Time (min): | 30        | Cycle Time (min): | 90     |
| Diameter:         | 40        | Length:           | 120    |
| Shape:            | round     |                   |        |

### Operation 3: Milling

|                   |             |                   |            |
|-------------------|-------------|-------------------|------------|
| Material:         | steel       | Machine:          | 3 axis CNC |
| Man Hours/Unit:   | 150         | Duty Category:    | heavy      |
| Setup Time (min): | 3           | Cycle Time (min): | 45         |
| Length:           | 200         | Breadth:          | 60         |
| Height:           | 50          |                   |            |
| Shape:            | rectangular |                   |            |

Operation 4: Grinding

|                   |             |                   |                              |
|-------------------|-------------|-------------------|------------------------------|
| Material:         | steel       | Machine:          | Special Purpose Machine(SPM) |
| Man Hours/Unit:   | 4000        | Duty Category:    | heavy                        |
| Setup Time (min): | 35          | Cycle Time (min): | 180                          |
| Diameter:         | 25          | Length:           | 80                           |
| Shape:            | rectangular |                   |                              |

Operation 5: Boring

|                   |             |                   |       |
|-------------------|-------------|-------------------|-------|
| Material:         | titanium    | Machine:          | CNC   |
| Man Hours/Unit:   | 3200        | Duty Category:    | heavy |
| Setup Time (min): | 50          | Cycle Time (min): | 210   |
| Diameter:         | 18          | Length:           | 90    |
| Shape:            | rectangular |                   |       |

Operation 6: Rubber press

|                   |             |                   |                                  |
|-------------------|-------------|-------------------|----------------------------------|
| Material:         | steel       | Machine:          | Small (less than 650 MMX 650 MM) |
| Man Hours/Unit:   | 2800        | Duty Category:    | heavy                            |
| Setup Time (min): | 40          | Cycle Time (min): | 300                              |
| Diameter:         | 150         | Length:           | 400                              |
| Shape:            | rectangular |                   |                                  |

Operation 7: Rubber press

|                   |             |                   |                                   |
|-------------------|-------------|-------------------|-----------------------------------|
| Material:         | aluminium   | Machine:          | Medium (more than 650 MMX 650 ... |
| Man Hours/Unit:   | 3500        | Duty Category:    | medium                            |
| Setup Time (min): | 55          | Cycle Time (min): | 280                               |
| Diameter:         | 220         | Length:           | 600                               |
| Shape:            | rectangular |                   |                                   |

## Operation 8: Drilling

|                   |             |                   |                  |
|-------------------|-------------|-------------------|------------------|
| Material:         | aluminium   | Machine:          | Drilling machine |
| Man Hours/Unit:   | 1200        | Duty Category:    | heavy            |
| Setup Time (min): | 15          | Cycle Time (min): | 60               |
| Diameter:         | 12          | Length:           | 50               |
| Shape:            | rectangular |                   |                  |

## Operation 9: Turning

|                   |             |                   |       |
|-------------------|-------------|-------------------|-------|
| Material:         | steel       | Machine:          | CNC   |
| Man Hours/Unit:   | 2600        | Duty Category:    | heavy |
| Setup Time (min): | 28          | Cycle Time (min): | 110   |
| Diameter:         | 55          | Length:           | 180   |
| Shape:            | rectangular |                   |       |

## Operation 10: Milling

|                   |             |                   |            |
|-------------------|-------------|-------------------|------------|
| Material:         | titanium    | Machine:          | 5 axis cnc |
| Man Hours/Unit:   | 240         | Duty Category:    | heavy      |
| Setup Time (min): | 5           | Cycle Time (min): | 60         |
| Length:           | 300         | Breadth:          | 70         |
| Height:           | 65          |                   |            |
| Shape:            | rectangular |                   |            |

## Operation 11: Grinding

|                   |             |                   |                              |
|-------------------|-------------|-------------------|------------------------------|
| Material:         | steel       | Machine:          | Special Purpose Machine(SPM) |
| Man Hours/Unit:   | 2900        | Duty Category:    | heavy                        |
| Setup Time (min): | 32          | Cycle Time (min): | 160                          |
| Diameter:         | 30          | Length:           | 100                          |
| Shape:            | rectangular |                   |                              |

## Operation 12: Boring

|                   |             |                   |       |
|-------------------|-------------|-------------------|-------|
| Material:         | steel       | Machine:          | CNC   |
| Man Hours/Unit:   | 3300        | Duty Category:    | heavy |
| Setup Time (min): | 48          | Cycle Time (min): | 200   |
| Diameter:         | 20          | Length:           | 85    |
| Shape:            | rectangular |                   |       |

## Operation 13: Rubber press

|                   |             |                   |                                   |
|-------------------|-------------|-------------------|-----------------------------------|
| Material:         | steel       | Machine:          | Medium (more than 650 MMX 650 ... |
| Man Hours/Unit:   | 4100        | Duty Category:    | heavy                             |
| Setup Time (min): | 58          | Cycle Time (min): | 320                               |
| Diameter:         | 250         | Length:           | 750                               |
| Shape:            | rectangular |                   |                                   |

## Operation 14: Drilling

|                   |             |                   |                  |
|-------------------|-------------|-------------------|------------------|
| Material:         | steel       | Machine:          | Drilling machine |
| Man Hours/Unit:   | 1900        | Duty Category:    | heavy            |
| Setup Time (min): | 22          | Cycle Time (min): | 95               |
| Diameter:         | 18          | Length:           | 60               |
| Shape:            | rectangular |                   |                  |

## Operation 15: Turning

|                   |             |                   |              |
|-------------------|-------------|-------------------|--------------|
| Material:         | aluminium   | Machine:          | Conventional |
| Man Hours/Unit:   | 1500        | Duty Category:    | light        |
| Setup Time (min): | 18          | Cycle Time (min): | 70           |
| Diameter:         | 35          | Length:           | 140          |
| Shape:            | rectangular |                   |              |

## Operation 16: Milling

|                   |       |                   |        |
|-------------------|-------|-------------------|--------|
| Material:         | steel | Machine:          | CNC    |
| Man Hours/Unit:   | 2700  | Duty Category:    | medium |
| Setup Time (min): | 40    | Cycle Time (min): | 130    |
| Length:           | 160   | Breadth:          | 45     |
| Height:           | 40    |                   |        |
| Shape:            | round |                   |        |

## Operation 17: Grinding

|                   |             |                   |                              |
|-------------------|-------------|-------------------|------------------------------|
| Material:         | titanium    | Machine:          | Special Purpose Machine(SPM) |
| Man Hours/Unit:   | 3600        | Duty Category:    | heavy                        |
| Setup Time (min): | 38          | Cycle Time (min): | 190                          |
| Diameter:         | 28          | Length:           | 95                           |
| Shape:            | rectangular |                   |                              |

## Operation 18: Boring

|                   |             |                   |       |
|-------------------|-------------|-------------------|-------|
| Material:         | steel       | Machine:          | CNC   |
| Man Hours/Unit:   | 3900        | Duty Category:    | heavy |
| Setup Time (min): | 52          | Cycle Time (min): | 230   |
| Diameter:         | 24          | Length:           | 110   |
| Shape:            | rectangular |                   |       |

## Operation 19: Rubber press

|                   |             |                   |                                  |
|-------------------|-------------|-------------------|----------------------------------|
| Material:         | steel       | Machine:          | Small (less than 650 MMX 650 MM) |
| Man Hours/Unit:   | 3000        | Duty Category:    | heavy                            |
| Setup Time (min): | 42          | Cycle Time (min): | 260                              |
| Diameter:         | 180         | Length:           | 500                              |
| Shape:            | rectangular |                   |                                  |

Operation 20: Rubber press

|                   |             |                   |                                   |
|-------------------|-------------|-------------------|-----------------------------------|
| Material:         | aluminium   | Machine:          | Medium (more than 650 MMX 650 ... |
| Man Hours/Unit:   | 4500        | Duty Category:    | medium                            |
| Setup Time (min): | 60          | Cycle Time (min): | 340                               |
| Diameter:         | 300         | Length:           | 900                               |
| Shape:            | rectangular |                   |                                   |